FX Ambiguity and Safe-Haven Reserve Allocation

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Abstract

This project examines whether FX ambiguity influences central-bank reserve allocations. Using the monthly cross-sectional ambiguity measure AMBt from Asano (2025), we construct a quarterly global ambiguity index from daily FX spot returns and merge it with IMF COFER reserve data. We focus on the shares of two major safe-haven currencies, USD and CHF, and evaluate whether ambiguity shocks lead to higher safe-haven holdings. Our results show a weak negative correlation between FX ambiguity and safe-haven reserve shares, indicating that central banks do not increase their USD or CHF allocations during periods of elevated ambiguity. These findings contrast with private-market behavior observed in the literature and suggest that reserve managers respond differently to uncertainty.

1. Introduction

Understanding the role of uncertainty in international financial markets is increasingly important for both investors and policymakers. In foreign exchange (FX) markets, ambiguity, defined as uncertainty about the underlying probability distribution of returns, can alter risk-taking behavior in ways that standard volatility-based data cannot fully capture, as Asano, Izhakian, and coauthors (2025) demonstrates that FX ambiguity influences the performance of currency carry trades.

Unlike individual and private market traders, central banks do not operate with short-term profit motives. Their reserve management decisions emphasize safety, liquidity, and the global acceptability of the assets they hold. This motivates our central question: Does higher FX ambiguity lead global reserves arrangement toward safe-haven currencies such as the U.S. dollar

and Swiss franc? To address this, we construct a quarterly global ambiguity index based on daily FX spot returns and the Asano's observable $\mho^2$ methodology, and we combine it with quarterly global reserve composition data from the IMF COFER database. Our main result indicates that global safe-haven reserve shares exhibit a negative weak correlation with FX ambiguity. In other words, when ambiguity rises, central banks do not appear to increase their holdings of USD or CHF. This suggests that the ambiguity–reserve relationship differs from the behavior observed in private-sector carry trades, highlighting the unique constraints and objectives that shape official reserve management.

2. Theoretical foundation and methodology

2.1 Ambiguity Estimation

Izhakian's (2020) ambiguity framework provides a formal way to measure second-order uncertainty – uncertainty not only about outcomes themselves but about the probability model governing those outcomes. Ambiguity arises when the decision-maker cannot identify a single probability model and instead considers multiple plausible distributions. This idea is captured by the ambiguity measure:

$$\mho^2(h) = \int_X E[\varphi_h(x)] * Var[\varphi_h(x)]\, dx$$

Where $\varphi_h(x) = P(\{\omega \in \Omega \,|h(\omega) = x\})$ represent the marginal probability density function of outcome h (where h is a random variable) . $E[\varphi_h(x)] = \int_\rho \varphi_h(x)\, d\xi$ indicates the expectation and $Var[\varphi_h(x)] = \int_\rho (\varphi_h(x) - E[\varphi_h(x)])^2\, d\xi$ indicated the variance and uncertainty of the

probability of $x \in X$ over p where p represents a set of probability measures. The measure integrates this weighted variance across all outcomes, producing a scalar value that summarizes how unstable or uncertain the underlying distribution is.

While conceptually powerful, this framework has practical limitations. The set of subjective probability distributions required for computing $\varphi_h(x)$ is unobservable in real financial markets, and continuous densities are difficult to estimate accurately from finite daily return data. As a result, empirical studies - including the one replicated in this project - use a discretized version of this measure, AMB□, which approximates ambiguity using observable daily FX return distributions rather than the unobservable theoretical priors.

2.2 Empirical Ambiguity Index (AMB□)

To estimate monthly cross-sectional FX ambiguity, the method begins by computing daily spot returns for each currency pair, defined as:

$$r_{i,d} = s_{i,d} - s_{i,d-1}$$

Where $s_{i,d}$ is the log of the spot exchange rate of currency i on day d. For easier computation, we omit i and d and simply leave the formula with r once we capture the daily spot return for a currency pair. For each trading day, the realized return across K currency pairs is treated as a one prior distribution in the sense of Izhakian (2020). Over a typical month with 22 trading days, this produces 22 different daily return distributions—representing a set of investors' priors. If all

days shared the same distribution, there would be no ambiguity; variation across these daily distributions generates ambiguity.

To quantify this ambiguity, the method discretizes daily returns into 12 bins. The range from -2% to +2% is divided into 10 equal width bins (with increment of 0.4%), and two additional bins capture returns below -2% and above +2%. For each bin B^j , the probability $P(B^j)$ of falling into that interval is computed daily using a normal distribution with day-specific mean μ_d and standard deviation σ_d . μ_d is calculated as the arithmetic average of returns r on day d for all currency pairs represented. $\sigma^2 = E(x^2) - E(x)^2$ is also known as the variance can simply be calculated once the mean has veen calculated. The monthly ambiguity index AMB□ is then obtained by calculating, for each bin, the mean of the daily probabilities $E[P(B^j)]$ and the variance of those probabilities $Var[P(B^j)]$, and summing their product across all bins:

$$
\begin{aligned}
& \mathrm{AMB_t} \\
= & \sum_{j=0}^{11} E[P(B^j)]\mathrm{Var}[P(B^j)] \\
= & E[\Phi(r^0;\mu,\sigma)]\mathrm{Var}[\Phi(r^0;\mu,\sigma)] \\
+ & \sum_{j=1}^{10} E[\Phi(r^j;\mu,\sigma)-\Phi(r^{j-1};\mu,\sigma)]\mathrm{Var}[\Phi(r^j;\mu,\sigma)-\Phi(r^{j-1};\mu,\sigma)] \\
+ & E[1-\Phi(r^{10};\mu,\sigma)]\mathrm{Var}[1-\Phi(r^{10};\mu,\sigma)].
\end{aligned}
$$

Where $\Phi(.)$ denotes the cumulative normal distribution of each independent currency. In this case the 12 equally divided bins are are labeled into $r^0, r^1, \ldots, r^{11}$. The lowest bin with bound $(-\infty, -2\%)$ is calculated as $P(B^0) = \Phi(r^0, \mu, \sigma)$. The middle 10 bins with 0.4% increments (j=1 to 10) are calculated as the probability mass between two adjacent cutoff:

$P(B^j) = \Phi(r^j; \mu, \sigma) - \Phi(r^{j-1}; \mu, \sigma)$. The top bin with bound $(2\%, \infty)$ is calculated as: $P(B^{11}) = 1 - \Phi(r^{10}; \mu, \sigma)$. For each month, the mean of these probabilities $E[P(B^j)]$ measures how often the return falls into the bin on average. The variance $Var[P(B^j)]$ measures how much the probability fluctuates and how much uncertainty across days.

This discretized formula is the empirical counterpart of Izhakian's theoretical ambiguity measure. High AMB□ values indicate that the daily return distributions differ substantially across days in the month, meaning the shape of the FX return distribution is unstable. This reflects high model uncertainty – the core idea behind ambiguity.

2.3 Hypothesis

We aim to test that higher foreign exchange (FX) ambiguity, distinct from volatility, induces central banks to tilt their reserve allocations toward "safe-haven" currencies such as USD and CHF.

3. Methodology

Our empirical strategy combines high-frequency FX data with quarterly global reserve composition. On the FX side, we download daily USD-based spot rates for nine major currency pairs (CHF, EUR, GBP, JPY, AUD, CAD, CNY, NOK, and SEK) from Yahoo Finance, covering 2015-11-12 to 2025-11-12. We work with closing prices and compute daily log returns as the first difference of the log spot rate series for each currency. These daily cross-sectional returns form the input to our ambiguity index.

We then implement the Asano method of FX ambiguity index in Python. For each calendar month, we group daily returns and compute, for each trading day, the cross-sectional mean and cross-sectional standard deviation across the nine currencies. Days with missing values or zero standard deviation are discarded. Using these day-specific moments, we map returns into twelve bins using fixed boundaries between -2% and +2%, with the two outer bins capturing tail events below -2% and above +2%. For each day, we transform returns into bin probabilities via the normal CDF and obtain a 12-element probability vector. Monthly ambiguity (AMB_t) is computed as the sum over bins of the product between the mean of daily bin probabilities and their variance across days, exactly matching the discretized version of the ambiguity measure discussed in the previous section. We then aggregate the monthly series to the quarterly level by simple averaging to obtain a quarterly FX Ambiguity Index (AMB_q) .

On the reserve side, we use the IMF COFER database. We first filter the COFER CSV to include only observations from 2016 onward and reshape the data from wide to long format by quarter. We restrict attention to entries where COUNTRY = “World”, FREQUENCY = “Quarterly”, and TYPE_OF_TRANSFORMATION = “Shares”. Within this subset, we extract the series “Claims in US dollar” and “Claims in Swiss franc”, which we rename USD_share and CHF_share, respectively. We then merge these quarter-level series with the quarterly ambiguity index using the quarter label (e.g., “2016-Q1”). From the merged dataset, we construct combined USD_share + CHF_share = Total_SAFE_Share.

Together, the construction of AMB_q from daily cross-currency returns and the merging with COFER based reserve shares provides a consistent framework to test our hypothesis that higher FX ambiguity is linked to shifts in global reserve allocation toward USD and CHF.

4. Result and Analysis

Preliminary empirical patterns suggest that FX ambiguity and global reserve allocation move on different time scales. The ambiguity index is highly volatile and responds immediately to market uncertainty, while reserve shares evolve gradually due to institutional inertia, multi-quarter decision cycles, and valuation effects.

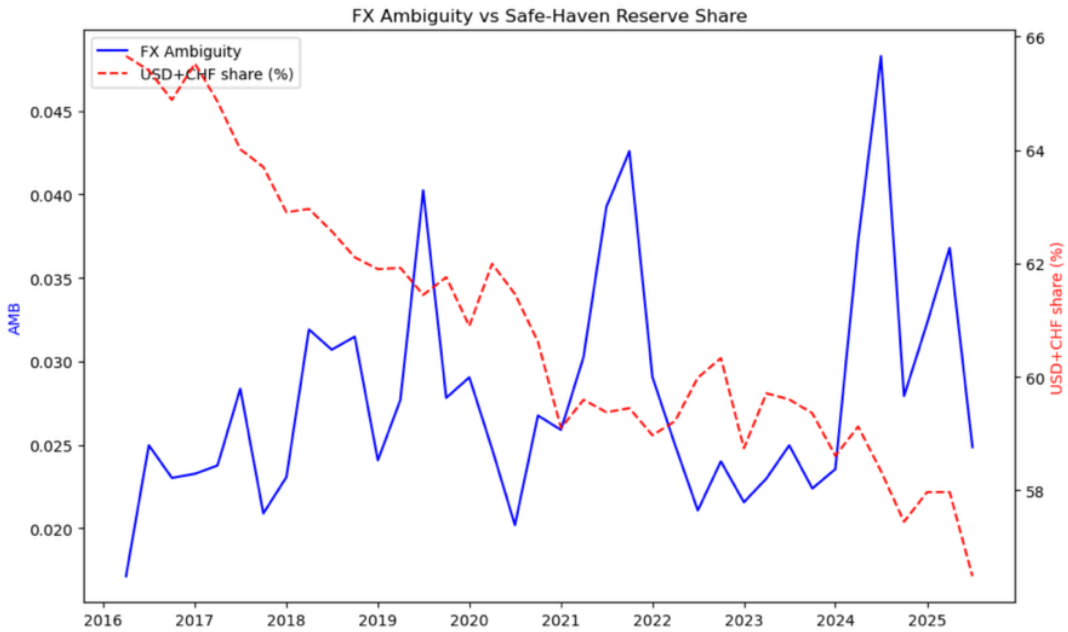

Image 1. This graph shows how FX quarterly Ambiguity and shares of two safe currencies changed over the period from first quarter of 2016 to first quarter of 2025. The red line represents the shares in percentage. The percent of shares generally decreased over time, while the FX Ambiguity fluctuated with a slightly increasing trend.

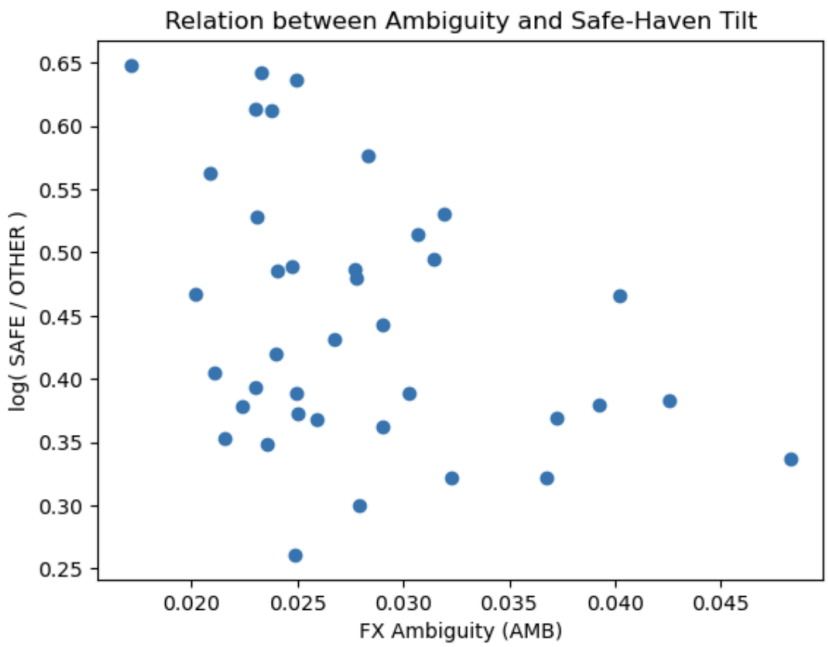

Image 2. Scatter plot showing a weak negative relationship between FX ambiguity (AMB) and the safe-haven reserve tilt, with substantial dispersion indicating limited sensitivity of reserve shares to short-term ambiguity.

The scatter plot indicates that the relationship between FX ambiguity (AMB) and the safe-haven reserve tilt is weak and not clearly directional. While there is a slight downward tendency – suggesting that higher ambiguity may be associated with a lower tilt toward safe-haven currencies – the points are broadly dispersed across the graph, and similar levels of ambiguity correspond to a wide range of safe-haven ratios. Overall, the wide dispersion of points suggests that central banks do not react strongly or consistently to short-term ambiguity shocks, and their reserve choices tend to move slowly over time rather than responding directly to month-to-month uncertainty in FX markets.

A simple correlation analysis shows a moderate negative relationship between ambiguity and safe-haven reserve shares (AMB vs SAFE_share = -0.36; AMB vs USD_share = -0.35). The

CHF_share correlation is weaker (-0.14), reflecting that the Swiss franc acts primarily as a crisis-specific safe haven rather than a structural reserve currency.

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                            OLS Regression Results
==============================================================================
Dep. Variable:             SAFE_share   R-squared:                       0.019
Model:                            OLS   Adj. R-squared:                 -0.009
Method:                 Least Squares   F-statistic:                    0.6893
Date:                Thu, 13 Nov 2025   Prob (F-statistic):              0.412
Time:                        15:24:30   Log-Likelihood:                -36.505
No. Observations:                  37   AIC:                             77.01
Df Residuals:                      35   BIC:                             80.23
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.1294      0.467      0.277      0.784      -0.819       1.078
AMB          -13.4956     16.255     -0.830      0.412     -46.496      19.505
==============================================================================
Omnibus:                        0.103   Durbin-Watson:                   2.458
Prob(Omnibus):                  0.950   Jarque-Bera (JB):                0.313
Skew:                          -0.045   Prob(JB):                        0.855
Kurtosis:                       2.558   Cond. No.                         148.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
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Image 3. This table shows the result of OLS regression.

These results imply that contemporaneous movements in ambiguity do not translate directly into reserve shifts. Since central banks adjust portfolios infrequently and often in response to sustained rather than temporary uncertainty, the relevant transmission channel is likely lagged ambiguity rather than same-period ambiguity. Visual inspection reinforces this: spikes in ambiguity rarely coincide precisely with changes in reserve shares, suggesting delayed or nonlinear effects. Overall, the constructed dataset is internally consistent and provides a solid foundation for the next stage of analysis, which will estimate whether lagged ambiguity significantly predicts future shifts in USD and CHF reserve shares after controlling for volatility, interest differentials, and valuation pressures.

5. Conclusion

This project investigated whether foreign exchange ambiguity - distinct from volatility, measured using the Asano et al. (2025) methodology, influences global central-bank reserve allocation toward safe-haven currencies. By constructing a quarterly FX Ambiguity measure from daily cross-currency returns and combining it with IMF COFER data on USD and CHF reserve shares, we find little evidence that rising ambiguity leads central banks to increase safe-haven holdings. Instead, the relationship is weakly negative, suggesting that reserve managers respond differently to uncertainty than private investors engaged in currency carry trades. Our results highlight that central-bank reserve allocation is driven more by structural, long-horizon considerations than by short-term ambiguity shocks in FX markets.

References:

1. Asano, Takao, et al. "Global Foreign Exchange Volatility, Ambiguity, and Currency Carry Trades." *Journal of Banking & Finance*, vol. 178, 107508, 2025, https://doi.org/10.1016/j.jbankfin.2025.107508.
2. Izhakian, Yehuda. "A Theoretical Foundation of Ambiguity Measurement." *Journal of Economic Theory*, vol. 187, 105001, 2020, https://doi.org/10.1016/j.jet.2020.105001.